

MICHAEL PEARCE

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Reed College, Department of Mathematics and Statistics
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CURRENT APPOINTMENT

Assistant Professor of Statistics (tenure-track)

Reed College, Department of Mathematics and Statistics

2023–Present

EDUCATION

University of Washington, Seattle, WA

2018–2023

Ph.D., Statistics (*Statistics in the Social Sciences* track)

Advisor: Elena A. Erosheva

Dissertation: Methods for the Statistical Analysis of Preferences, with Applications to Social Science Data

St. Olaf College, Northfield, MN

2013–2017

B.A., Mathematics (Statistics concentration)

Summa cum laude, Phi Beta Kappa

PUBLICATIONS

Refereed Articles

8. **Pearce, M.** and Zhou, S. (2026) “RaCE: a Rank-Clustering Estimation Method for Network Meta-Analysis.” *Research Synthesis Methods* 17.2: 314–331.
7. **Pearce, M.** and Erosheva, E.A. (2025) “Bayesian Rank-Clustering” *Psychometrika* 90.3: 904–931.
6. **Pearce, M.** and Perlman, M. (2025) “Estimating the ratio of means in a zero-inflated Poisson mixture model” *Stats* 8.3: 55.
5. **Pearce, M.** and Erosheva, E.A. (2025) “Modeling Preferences: A Bayesian Mixture of Finite Mixtures for Rankings and Ratings” *Journal of the American Statistical Association* 120.551: 1621–1632.
4. Gallo, S.A., **Pearce, M.**, Lee, C.J., and Erosheva, E.A. (2023) “A new approach to peer review assessments: Score, then rank.” *Research Integrity and Peer Review* 8.10.
3. **Pearce, M.** and Erosheva, E.A. (2022) “A unified statistical learning model for rankings and scores with application to grant panel review.” *Journal of Machine Learning Research* 23.210: 1–33.
2. **Pearce, M.** and Raftery, A.E. (2021) “Probabilistic forecasting of maximum human lifespan by 2100 using Bayesian population projections.” *Demographic Research* 44.52: 1271–1294.
1. **Pearce, M.***, Sparrow, Z. *, Mabote, T. R., and Sanchez-Gonzalez, R. (2020) “stoBEST: An efficient methodology for increased spatial resolution in two-component molecular tagging velocimetry.” *Measurement Science and Technology* 32.3: 035302. (* indicates authors contributed equally.)

Preprints and Other Scholarly Works

3. **Pearce, M.** and Erosheva, E.A. (2022) “On the validity of bootstrap uncertainty estimates in the Mallows-Binomial model.” *arXiv preprint arXiv:2206.12365*.
2. **Pearce, M.** and Raftery, A.E. (2021) “Will this be a record-breaking century for human longevity?” *Significance*.
1. **Pearce, M.** and Raftery, A.E. (2021) “The maximum human life span will likely increase this century, but not by more than a decade” *The Conversation*.

SOFTWARE

RaCE.NMA: Rank-Clustering Estimation of treatments in a Network Meta-Analysis. R package available on CRAN. Vignettes available [here](#).

rankclust: Fit a Bayesian, Rank-Clustered Bradley-Terry-Luce Model to Ordinal Comparison Data. R package available on Github. Vignettes available [here](#).

rankrate: Statistical Tools for Preference Learning with Rankings and Ratings. R package available on CRAN. Vignettes available [here](#).

Peer Review with Rankings and Ratings. R Shiny application.

RESEARCH PRESENTATIONS

24. “Bayesian rank-clustering of treatments from analyses of clinical trials data” *Working Group on Model-Based Clustering*, Edmonton, AB, July 2026. (*invited session*)
23. “Bayesian rank-clustering, with applications to survey and election data” *ISBA World Meeting*, Nagoya, Japan, July 2026. (*contributed session*)
22. “Ranked-choice voting and mixture modeling: An application to the 2025 New York City Democratic primary” *Classification Society Annual Meeting*, Halifax, NS, June 2026. (*invited session*)
21. “Rank-clustering treatments in network meta-analyses of clinical trials data” *Oregon Chapter of the ASA Fall Meeting*, Portland, OR, November 2026. (*invited speaker*)
20. “Bayesian Rank-Clustering” *15th Scientific Meeting Classification and Data Analysis Group*, Naples, Italy, September 2025 (*invited session*)
19. “Rank-clustering treatments in network meta-analyses of clinical trials data” *Classification Society Annual Meeting*, Ottawa, ON, June 2025. (*contributed session*)
18. “Statistical Perspectives on Ranked-Choice Voting” *Statistics and Machine Learning in the Social Sciences Working Group*, Seattle, WA, February 2025. (*invited speaker*)
17. “Statistical Estimation with Ranked Choice Voting” *University of Washington*, Seattle, WA, October 2024. (*invited lecture in STAT 452*)
16. “Bayesian Rank-Clustering” *Washington State University Vancouver*, Vancouver, WA, October 2024. (*invited seminar*)
15. “Bayesian Rank-Clustering” *Penn State University; Health Science Department*, University Park, PA, October 2024. (*invited seminar*)
14. “Bayesian Rank-Clustering” *Reed College*, Portland, OR, September 2024. (*colloquium*)
13. “Broadening Access to Bayesian Statistics with Active Learning” *Joint Statistical Meetings*, Portland, OR, August 2024. (*contributed session*)
12. “Bayesian Rank Clustering” *ISBA World Meeting*, Venice, Italy, July 2024. (*poster*)
11. “Methods for the Statistical Analysis of Preferences, with Applications to Social Science Data” *Classification Society Annual Meeting*, Kelowna, BC, June 2024. (*Distinguished Dissertation Award session*).
10. “Bayesian Rank Clustering” *Center for Statistics in the Social Sciences 25th Anniversary Conference* Seattle, WA, May 2024. (*poster*)
9. “Weighted Mallows-Binomial Model for Rankings and Ratings” *Statistics and Machine Learning in the Social Sciences Working Group*, Seattle, WA, April 2024. (*invited speaker*)
8. “Rankings and ratings in peer review: A mixture of finite mixtures accounting for degree of reviewer leniency” *Statistics and Machine Learning in the Social Sciences Working Group*, Seattle, WA, November 2023. (*invited speaker*)
7. “Improving preference analysis: Joint models for ordinal and cardinal data” *International Meeting of the Psychometrics Society*, College Park, MD, July 2023. (*contributed session*)
6. “A Unified Statistical Learning Model for Rankings and Scores with Application to Grant Panel Review” *NeurIPS*, New Orleans, LA, December 2022. (*poster*)

5. “Fast Bayesian estimation for ranking models” *Joint Statistical Meetings*, Washington, DC, August 2022. (*contributed session*)
4. “Using ranking data for decision-making” *Joint Statistical Meetings*, Washington, DC, August 2022. (*contributed session chair*)
3. “Joint Bayesian inference for rankings and ratings under heterogeneous preferences” *ISBA World Meeting*, Montreal, QC, June 2022. (*poster*)
2. “Unified latent class modeling of scores and rankings applied to grant panel review” *Working Group on Model-Based Clustering*, Athens, Greece, October 2021. (*contributed session*)
1. “Unified latent class modeling of score and rank data applied to grant panel review” *Joint Statistical Meetings*, Seattle, WA, August 2021. (*contributed session*)

SELECTED MEDIA COVERAGE

- *Washington Post* “Want to add healthy years to your life? Here’s what new longevity research says.” (Oct. 11, 2021)
- *BBC News (Brazil)* “Por que cada vez mais pessoas estão vivendo até os 100 anos?” (July 11, 2022)
- *CNBC* “Researchers say the probability of living past 110 is on the rise — here’s what you can do to get there” (July 17, 2021)
- *Stats and Stories (Podcast)* “The Age of the Supercentenarian” (April 29, 2022)
- *Southern Weekly (China)* “What is the limit of human life span?” (Sept. 16, 2021)
- *Elemental (Medium)* “How Long Can Humans Really Live?” (July 15, 2021)
- *Gulf News* “Surviving up to 150: How long can a person live?” (July 12, 2021)
- *Indian Express* “Can a person live to age 124, 135 or 150? Some optimism, some caveats” (July 6, 2021)
- *The South African* “Rise of the supercentenarians: Today’s kids could live for 130 years” (July 4, 2021)
- *UW News* “How long can a person live? The 21st century may see a record-breaker” (July 1, 2021)

THESIS ADVISING

Conor Bekaert, <i>Towards a Weighted Joint Statistical Model for Rankings and Ratings</i>	2023–24
Quinn Hargrove, <i>Visualizations to Improve Ranked Data Analyses, with Applications to Board Game Data</i>	
Chance Addis, <i>Kingmaker: A Simulation Analysis of Strategic Voting</i>	2024–25
Emma Dunn, <i>A Bayesian Methodology for Forecasting Instant-Runoff Elections</i>	
Tory Norton, <i>How Many Mixtures? Frequentist and Bayesian Approaches to Inferring the Number of Latent Groups</i>	
Chris Li, <i>Evaluating Large Language Models for Data Science Tasks in R</i>	2025–26
Nathan Holland, <i>Spatial Statistics for Environmental Policy, with Application to Air Quality in Los Angeles</i>	
Deltas Shi, <i>Inferring Team Strength in Ultimate Frisbee: A Comparative Study of Ranking Systems</i>	

TEACHING

Reed College

Instructor, STAT 141 (Introduction to Probability and Statistics)	Fall 2023, Fall 2024
Instructor, STAT 243 (Statistical Learning)	Spring 2025, Fall 2025
Instructor, STAT 343 (Statistics Practicum)	Fall 2025
Instructor, STAT 346 (Bayesian Statistics)	Spring 2024, Spring 2026
Instructor, STAT 392 (Mathematical Statistics)	Spring 2025, Spring 2026

University of Washington

Instructor, CSSS 508 (Introduction to R for Social Scientists)	Autumn 2022, Spring 2023
Teaching Assistant, CSSS 589 (Multivariate Data Analysis for the Social Sciences)	Autumn 2021
Teaching Assistant, CSSS 594 (Statistics and Philosophy of Voting)	Autumn 2020
Teaching Assistant, STAT 220 (Statistical Reasoning)	Autumn 2019
Teaching Assistant, STAT 311 (Elements of Statistical Methods)	Autumn 2018, Winter 2019
Teaching Assistant, STAT 342 (Introduction to Probability and Mathematical Statistics)	Spring 2019
Teaching Assistant, STAT 528 (Applied Statistics Capstone)	Winter 2021, Winter 2022

Mentor, Directed Reading Program
Mentor, Washington eXperimental Mathematics Lab

7 quarters; Autumn 2020–Spring 2022
Autumn 2021

SERVICE

Reed College

Statistics program chair	2025–26
Library Board	2024–25, 2025–26
Undergraduate Research Committee	2023–24
Math and Statistics Department Colloquium organizer	2023–24

University of Washington

Pre-application review service reviewer	2023
Queer Union for (Bio)statistician Inclusion and Community affinity group founder	2022–23
Undergraduate statistics curriculum committee	2021–22
PhD program admissions reviewer	2020–21

Broader Scholarly Community

Bayesian Education Research and Practice (BERaP) ISBA section treasurer	2026–27
Scientific organizing committee for the Classification Society Annual Meeting	2026
Undergraduate Statistics Project Competition judge	2023, 2025
International Conference on Machine Learning workflow chair	2021
Manuscript review (various journals, e.g., <i>Annals of Applied Statistics</i> , <i>Journal of Statistics and Data Science Education</i> , <i>Journal of Open Research Software</i> , <i>Biomedicine Hub</i> , <i>Cogent Social Sciences</i>)	

HONORS AND AWARDS

Distinguished Dissertation Award (<i>The Classification Society</i>)	2024
Best Poster Award (<i>International Society for Bayesian Analysis World Meeting</i>)	2024
Travel Award (<i>International Society for Bayesian Analysis</i>)	2022, 2024
Scholar Award (<i>NeurIPS</i>)	2022
Dorothy M. Gilford Teaching Award (<i>University of Washington</i>)	2021

INDUSTRY EXPERIENCE

Boeing Research and Technology (<i>Applied Statistics PhD Intern</i>)	2019–2020
Deloitte LLC (<i>Analytics Consultant</i>)	2017–2018

PROFESSIONAL AFFILIATIONS

American Statistical Association
International Society for Bayesian Analysis
The Classification Society